EE3032 Introduction to VLSI Design
2024 Homework 4: Pre-layout SPICE Deliverable

Independent reconstruction and verification

13 October 2026

1 Requested scope and relation to the original assignment

The original Homework 4 asks for layout, extracted netlists and a pre-/post-layout comparison. This deliverable follows the user's explicit request to stop at the `.sp` stage. It provides a transistor-level schematic netlist and a runnable testbench. No layout, DRC, LVS, PEX or post-layout simulation is claimed.

The final Homework 3 netlist is copied without modifying transistor connections or dimensions. The copied file is `four_bit_multbased_adder.sp`; its hash and the new testbench hash are recorded in `statistics.json`. This makes the correspondence between the verified design and the Homework 4 deliverable explicit.

2 Function and interface

For bit i, define the equality signal $e_i = \overline{x_i \oplus y_i}$. The cell implements the two XNOR blocks and carry multiplexer in the assignment figure:

$$s_i = \overline{e_i \oplus c_i} = x_i \oplus y_i \oplus c_i, \qquad c_{i+1} = e_i y_i + \overline{e_i} c_i.$$

If $x_i = y_i$, the carry is y_i; if $x_i \neq y_i$, the carry is c_i. Four cells implement

$$\{\mathrm{COUT}, S_3, S_2, S_1, S_0\} = X + Y + \mathrm{CIN}.$$

The public subcircuit pin order is

`CIN X0 X1 X2 X3 Y0 Y1 Y2 Y3 S0 S1 S2 S3 COUT`.

The least significant bits are `X0`, `Y0`, `S0`. The testbench defines global rails `vdd` and `vss`.

3 Simulation and numerical checks

The external process file is the user's `D:/cic018.1`, TT corner, $V_{DD} = 1.8$ V and $T = 25°$C. The model is not included in the repository. The testbench preserves the instructor's PWL sources and CIN=0, simulating 0–9 ns with no extra output load. The sampling points are 1.8, 2.8, …, 8.8 ns, after each input vector has completed its transition.

Sample (ns)	X	Y	Expected $\{C_{out}, S\}$ (decimal)
1.8	1	5	6
2.8	15	13	28
3.8	11	9	20
4.8	4	8	12
5.8	3	7	10
6.8	12	14	26
7.8	2	15	17
8.8	6	10	16

The runnable verification records all five output voltages for all eight vectors. For an expected zero it checks $-0.02 \le V \le 0.2$ V; for an expected one it checks $1.6 \le V \le 1.82$ V. These tolerance bands allow small numerical overshoot while requiring output levels near both rails. The complete voltages are in `vectors.csv`; this numerical test does not use a GUI screenshot.

HSPICE's `power` includes dissipation and changes in stored energy. It is the negative sum of the powers absorbed by all independent sources, including the PWL input drivers. VDD-only power is measured separately as $-V(\text{vdd})I(\text{vvdd})$. The measured total energy must satisfy

$$E = \int_0^{9\,\text{ns}} P(t)\,dt = P_{\text{avg}}\,(9\,\text{ns}).$$

The file `statistics.json` records power, energy, pass counts, simulator version and source hashes. Homework 3 documents delay and exhaustive input checks for the same netlist. These are pre-layout results only.

Final netlist, independently rerun for this folder	Result
Transistor count	64
All widths / all lengths	$0.47/0.18\,\mu$m
$\sum_i W_i L_i$ (schematic WL metric)	$5.4144\,\mu\text{m}^2$
Total average HSPICE power	$69.558498\,\mu$W
Supply-delivered average power	$37.4328403\,\mu$W
Total energy over 9 ns	626.026482 fJ
Assigned-vector checks	8 / 8 pass

The power and energy values above use a 1 ps maximum timestep. Homework 3 separately records the original 0.1 ns-output-step grading testbench and its slightly different adaptive-step numerical results. Its full 512-case transient check applies to the identical transistor netlist supplied here.

4 Review of the existing Homework 4 answer

The supplied previous answer contains a 64-transistor schematic and extracted parasitic files. Its schematic was rerun as a separate DUT using the exact instructor stimulus. It also passed all 512 DC combinations of (X, Y, CIN) using the 0.2/1.6 V output criteria. This is evidence that the reference is functional under the tested conditions. The refined energy, delay and area comparison, using a 1 ps maximum timestep, is given in Homework 3. No unmeasured corner or load robustness is inferred.

The previous report reverses the pre-/post-layout labels: the extracted circuit with parasitics is the post-layout circuit. Its testbench simultaneously instantiates the schematic and the extracted design on a shared supply; `avg power` therefore combines both DUTs. The report's energy and layout area cannot be reused as measurements of the current design. The current testbench instantiates exactly one schematic DUT. The design comparison and selection are recorded in the Homework 3 report and statistics tables.

5 Reproduction and area distinction

Run `python reproduce.py` in this folder, or set `-hspice` and `-model` to the local simulator and process paths. Simulator raw outputs remain in `run_artifacts/`; the small statistics files are published. Licensed access to HSPICE and access to the external process file are required.

Homework 3 uses $A_{\text{WL}} = \sum_i W_i L_i$. Homework 4's physical layout area is the rectangle covering all layout materials. These measures are not interchangeable. Since this scope includes no physical layout, no physical area or post-layout delay/power is reported.